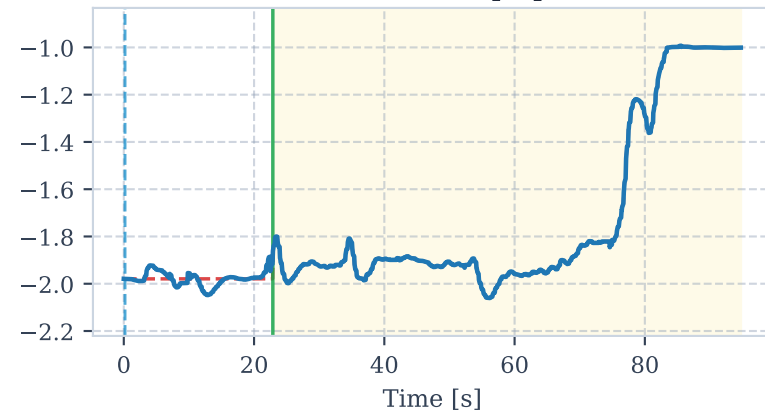


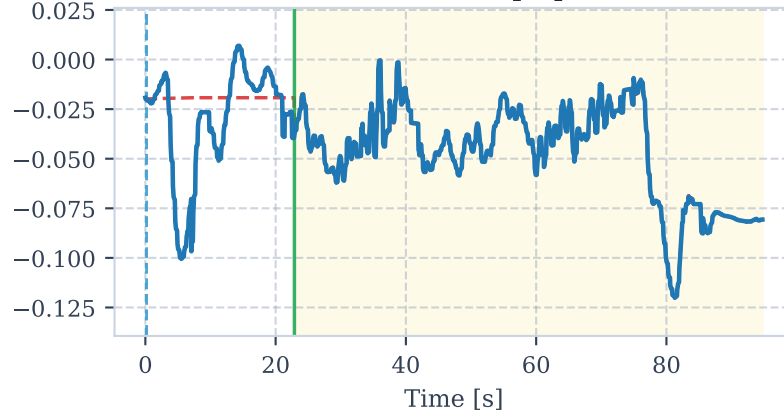
# Drone Position vs MPC optimal trajectory



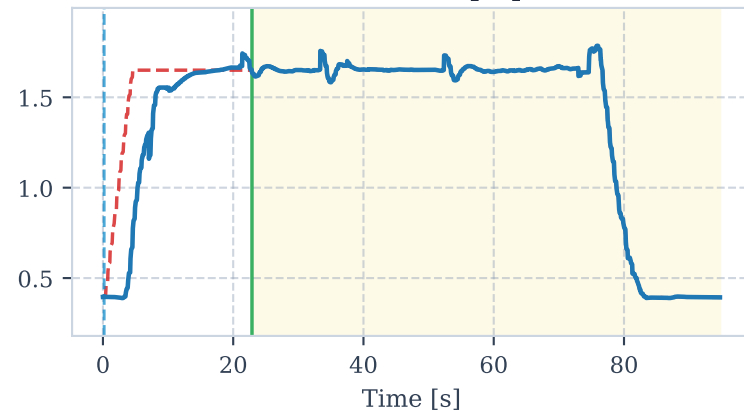
## Position X [m]



## Position Y [m]



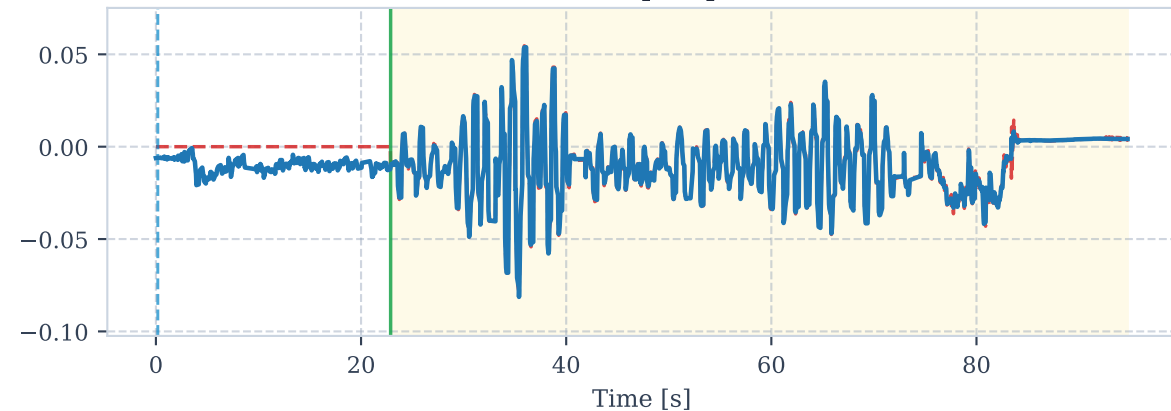
## Position Z [m]



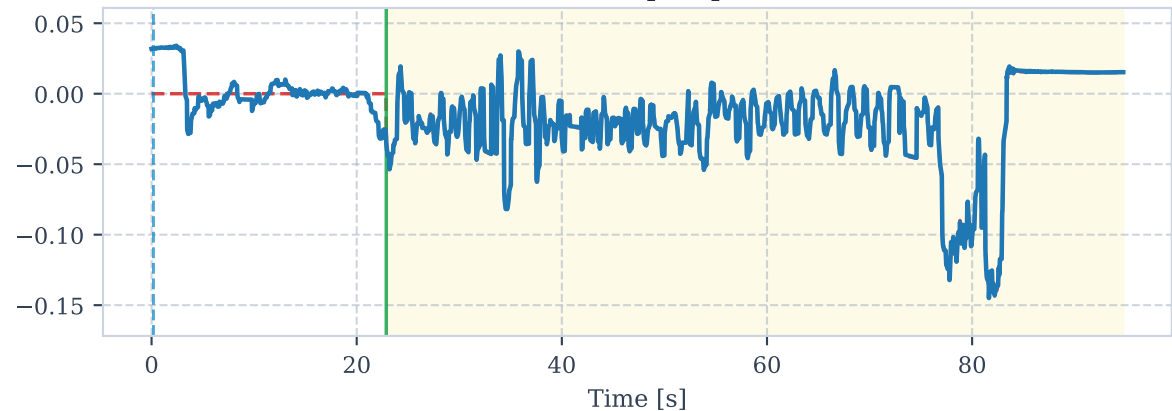
# Drone Orientation (Roll/Pitch)



## Roll [rad]



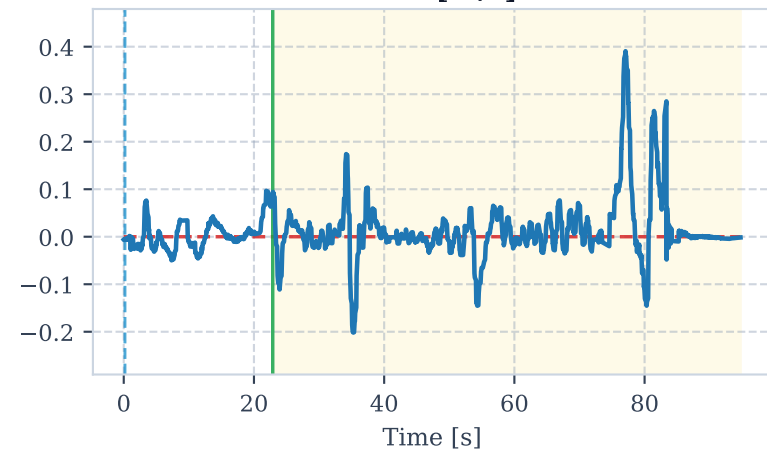
## Pitch [rad]



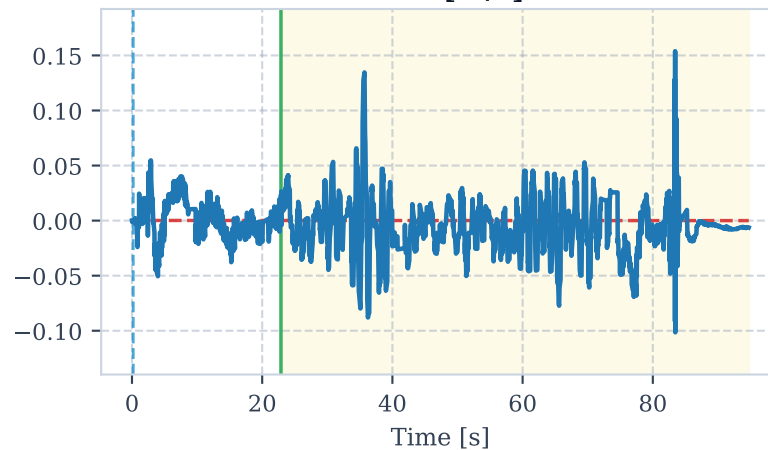
# Drone Velocities vs MPC Reference



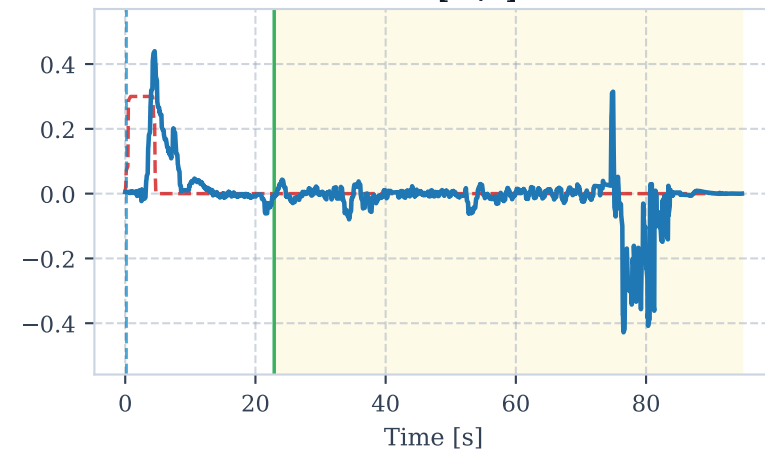
## Vel X [m/s]



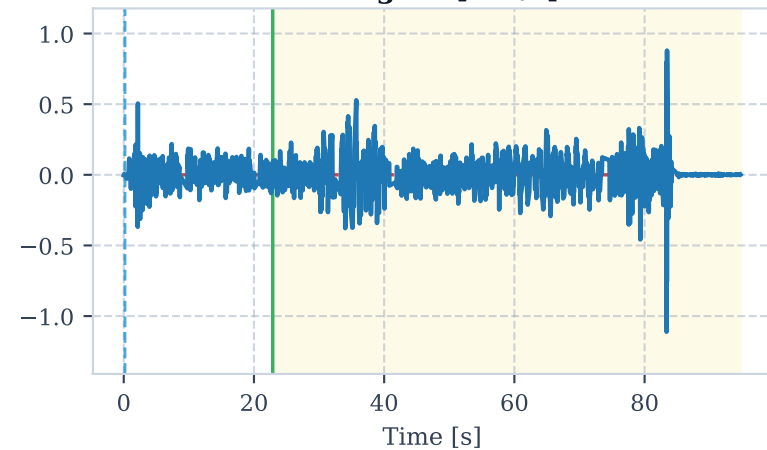
## Vel Y [m/s]



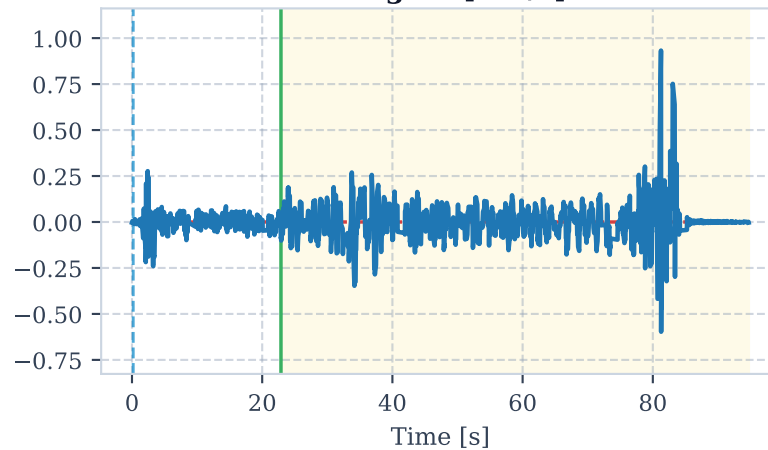
## Vel Z [m/s]



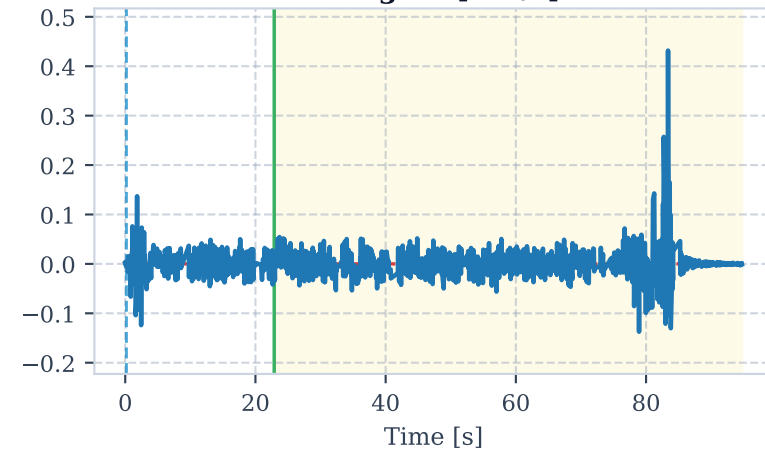
## Omega X [rad/s]



## Omega Y [rad/s]



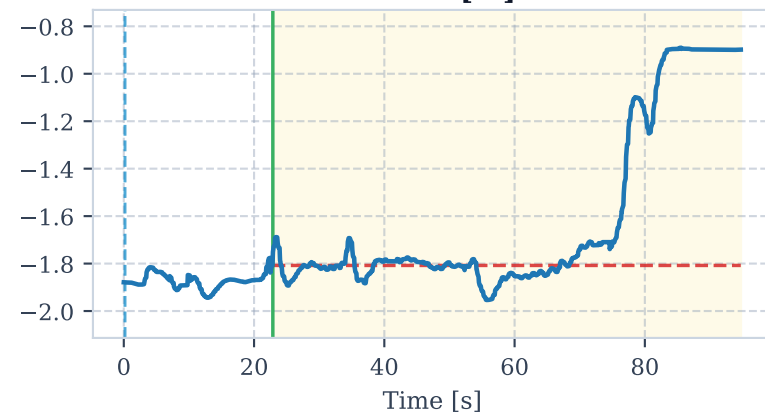
## Omega Z [rad/s]



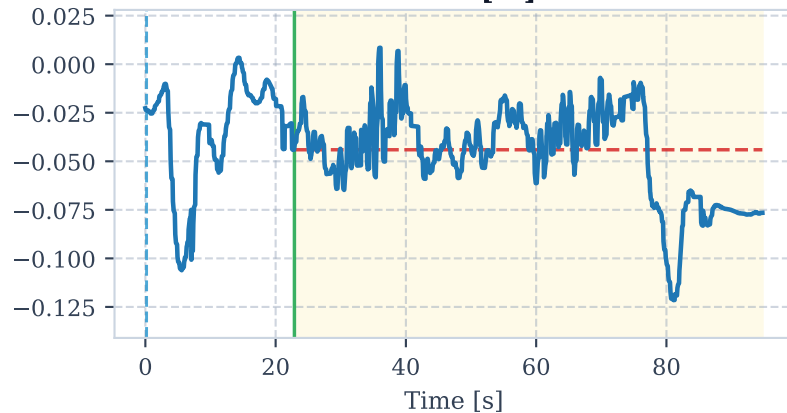
# Camera Cartesian Position vs Derived Target



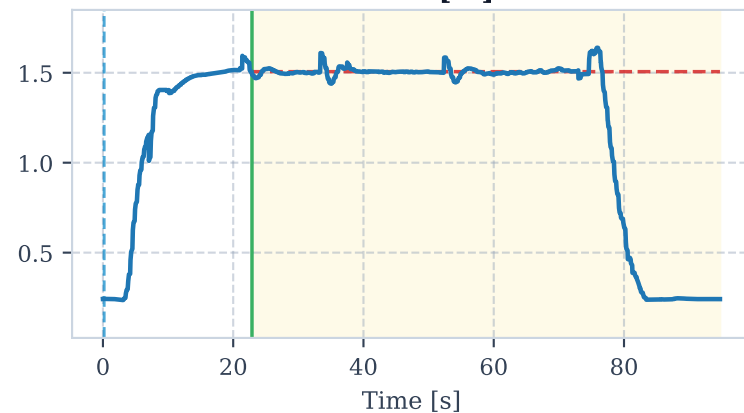
## Cam X [m]



## Cam Y [m]



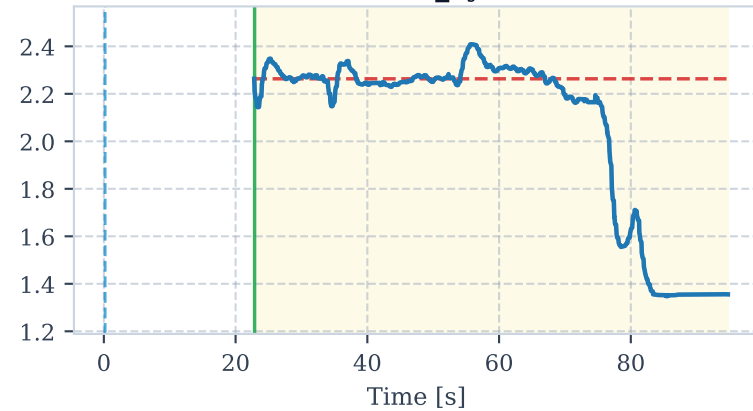
## Cam Z [m]



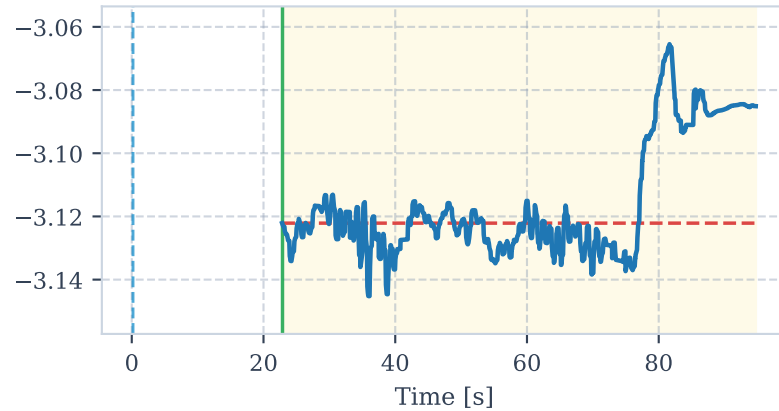
# Cylindrical PoV Tracking (World Frame)



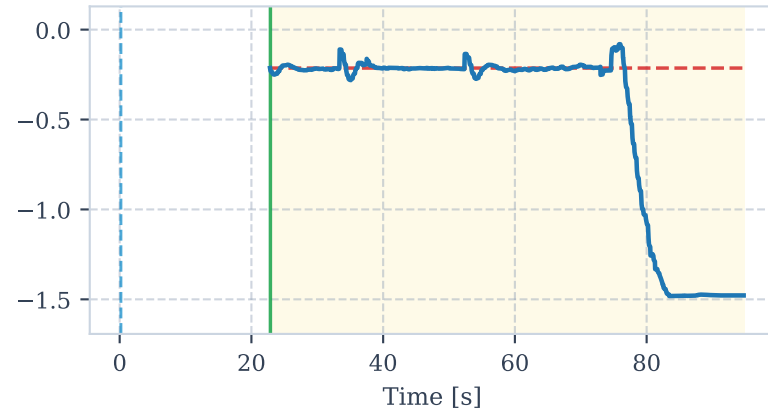
## Distance $r_{\text{cyl}}$ [m]



## Azimuth $\beta$ [rad]



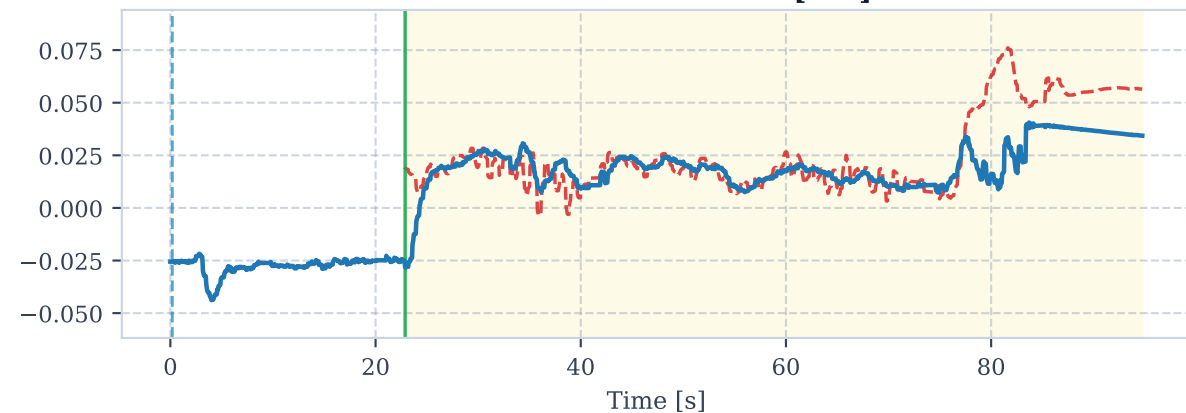
## Elevation $z$ [m]



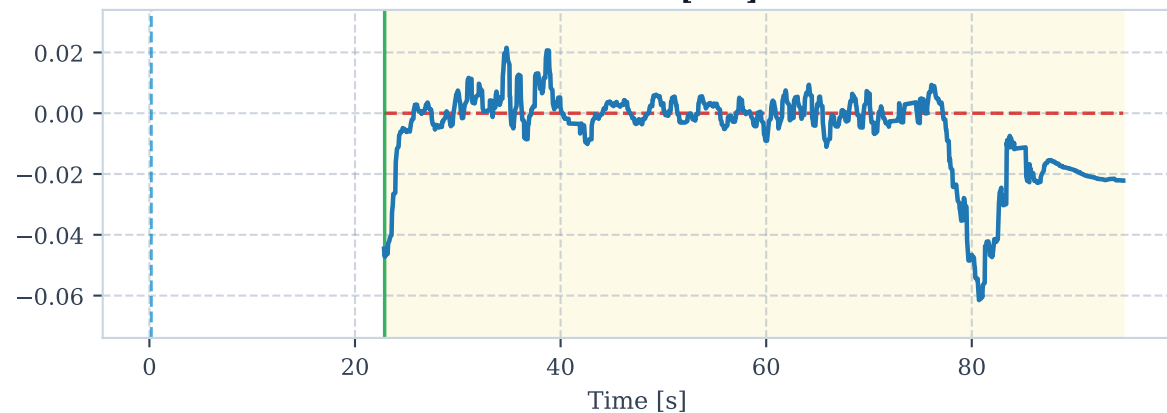
# Yaw Tracking: Drone Pointing Target



## Yaw Actual vs Desired [rad]



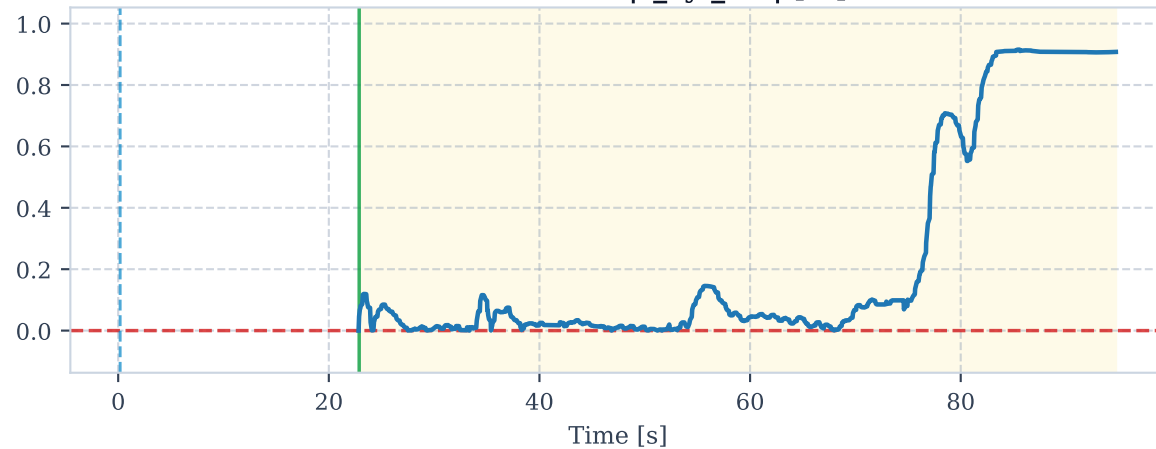
## Yaw Error [rad]



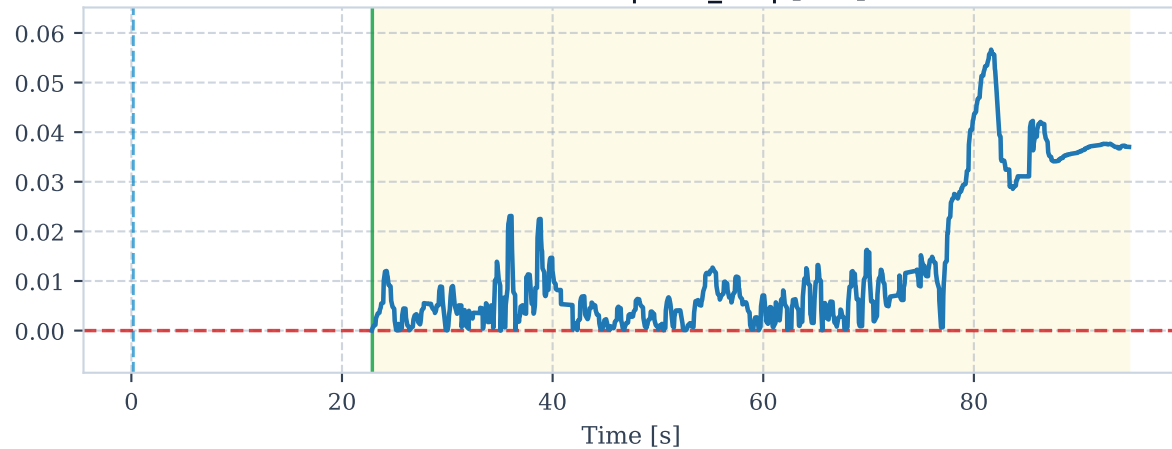
# Primary Tracking Errors (Cylindrical & Yaw)



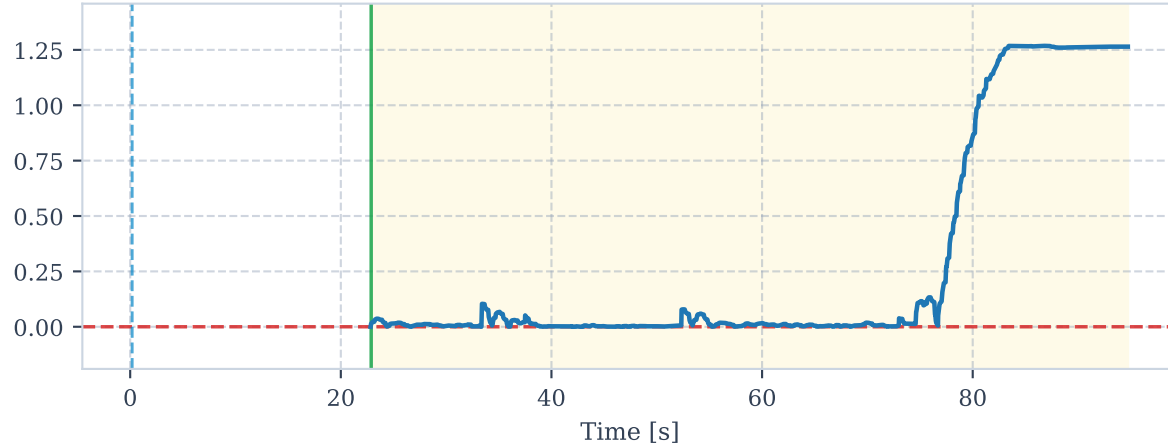
## Distance Error $|r_{cyl\_err}|$ [m]



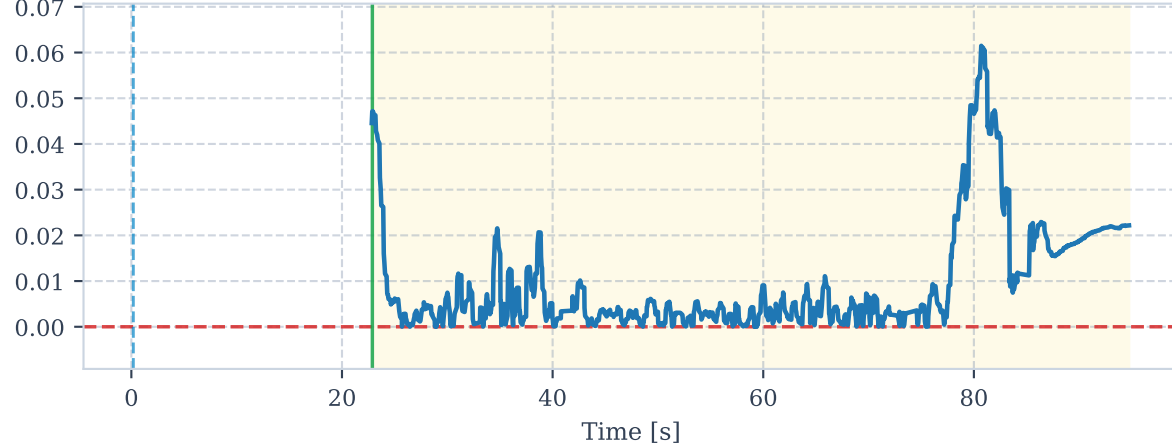
## Azimuth Error $|\beta_{err}|$ [rad]



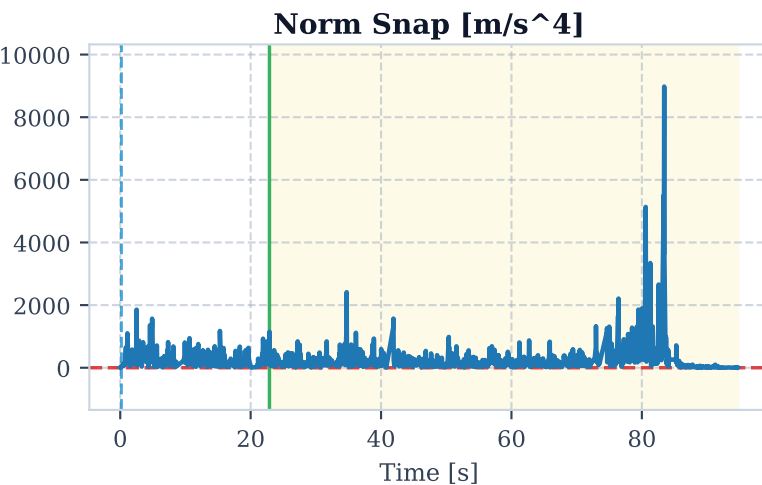
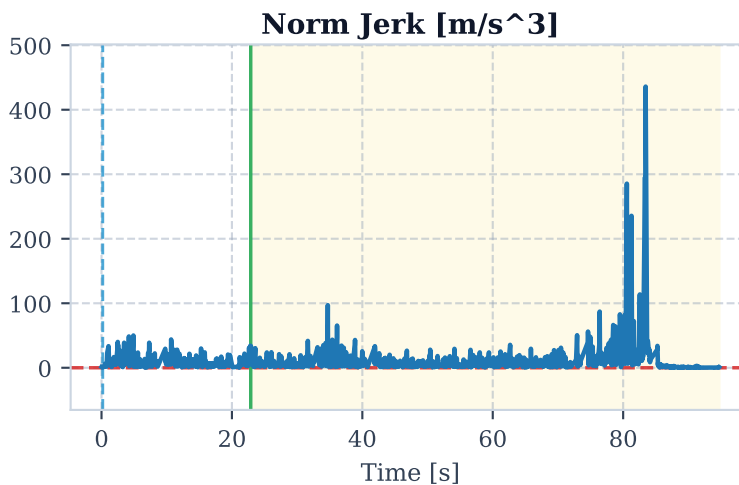
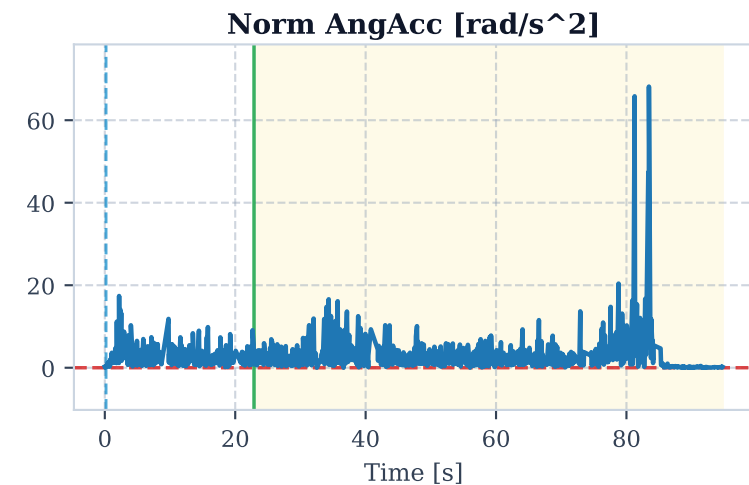
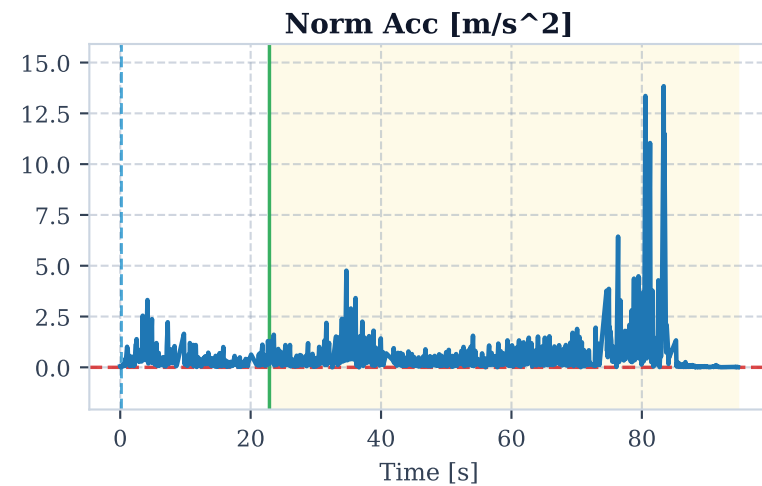
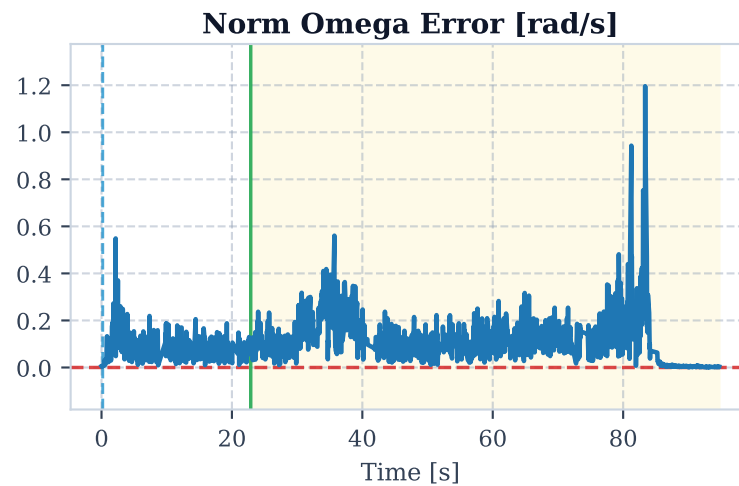
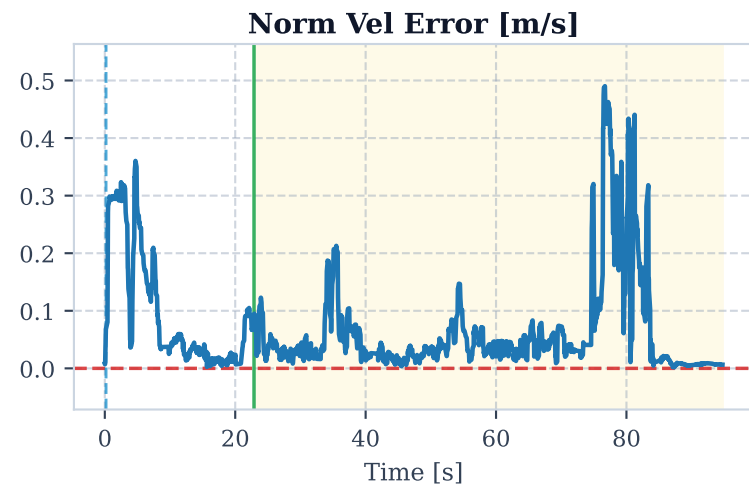
## Elevation Error $|z_{err}|$ [m]



## Yaw Error $|\text{yaw\_err}|$ [rad]



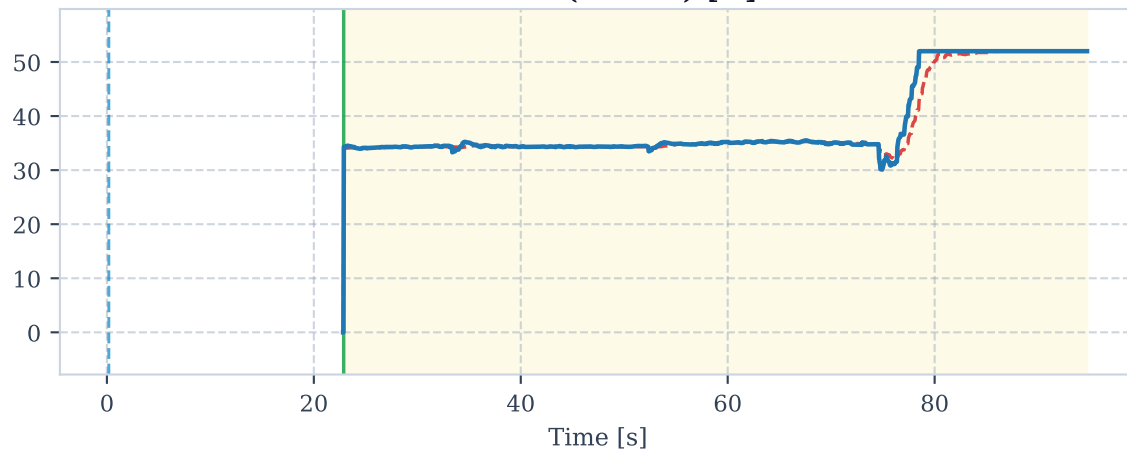
# Dynamic States Errors and Feedforward Derivatives



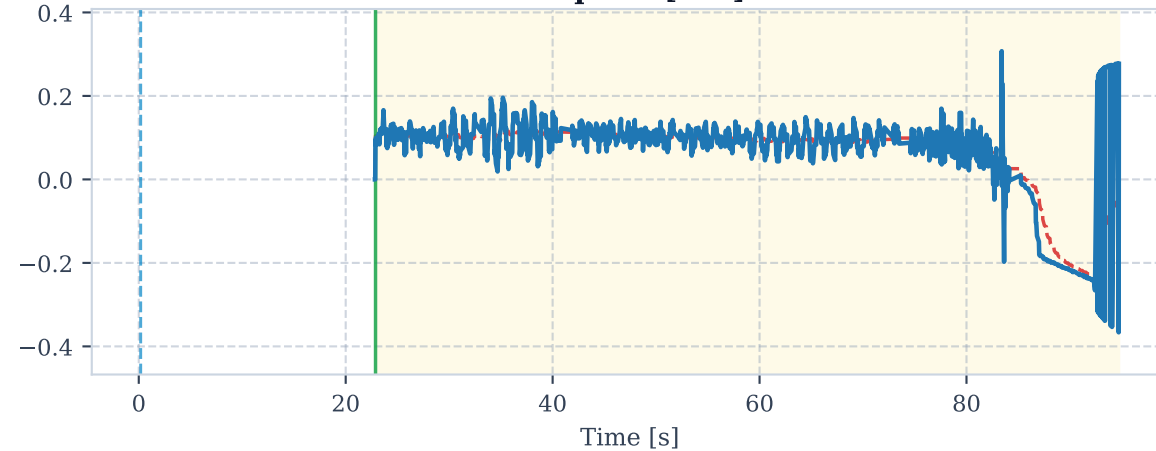
# Control Wrench (Hover Force = 20.24N)



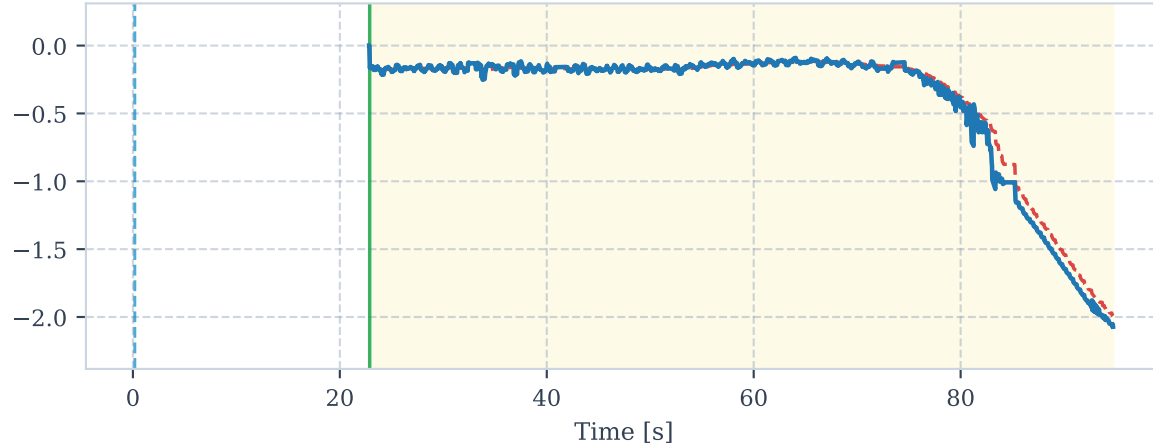
## Force Z (Thrust) [N]



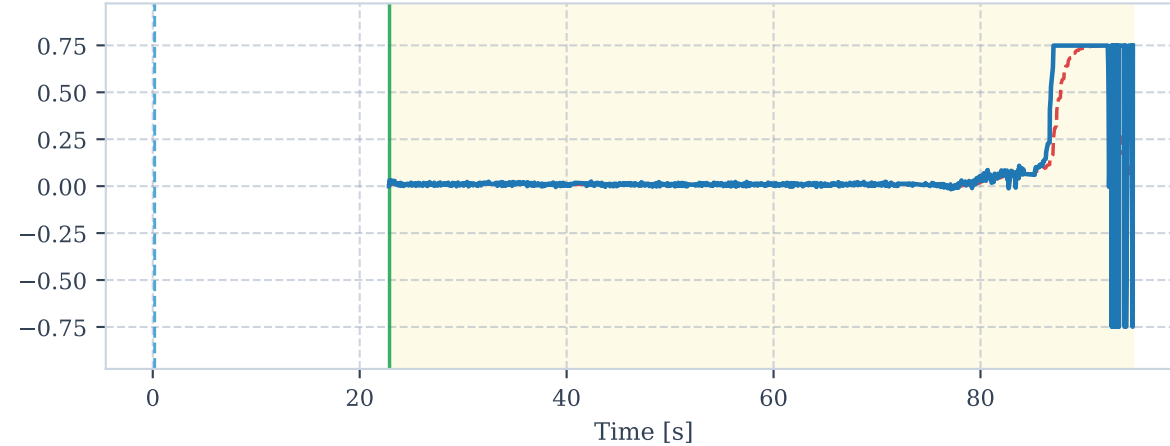
## Torque X [Nm]



## Torque Y [Nm]



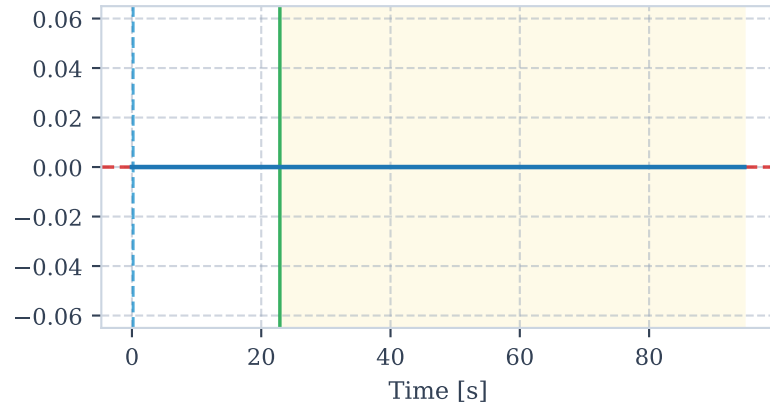
## Torque Z [Nm]



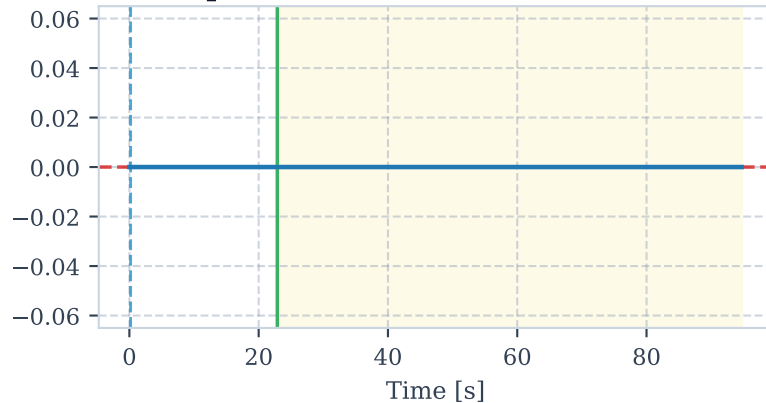
# Haptic APF Repulsive Forces (Safety Constraints)



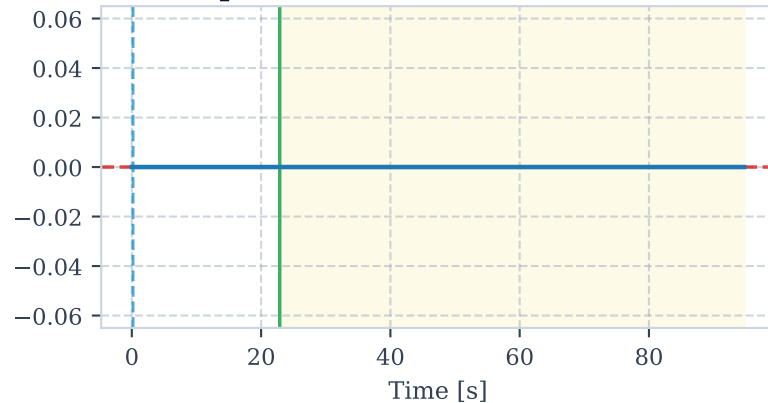
**Repulsive Force X (Distance limit  $r_{min}$ ) [N]**



**Repulsive Force Y (Horiz. FoV) [N]**



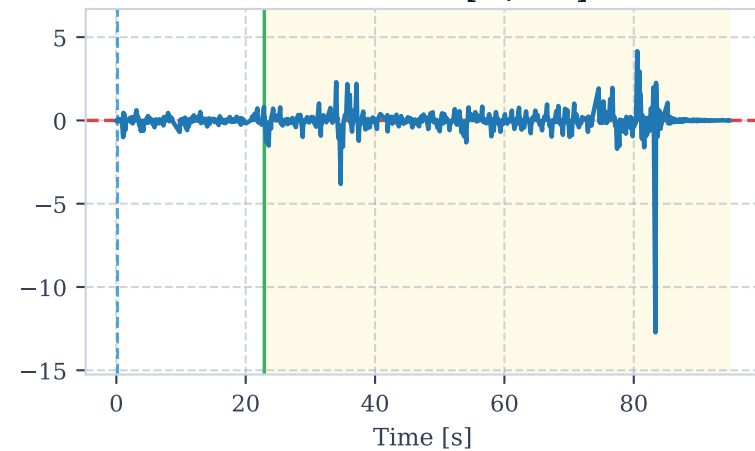
**Repulsive Force Z (Vert. FoV) [N]**



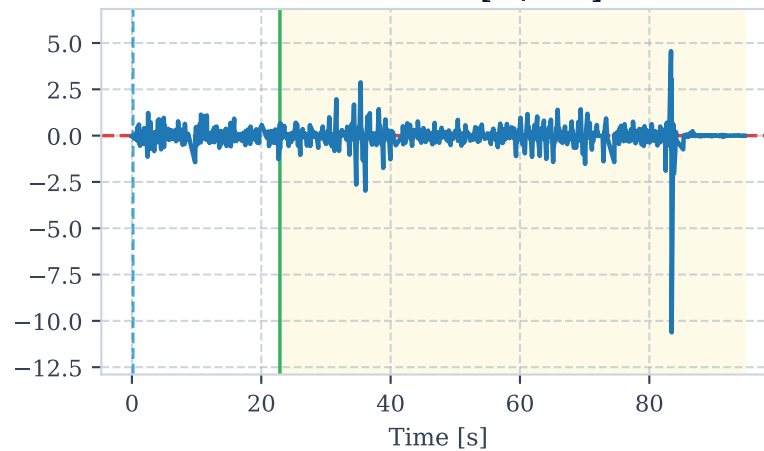
# Drone Linear and Angular Accelerations



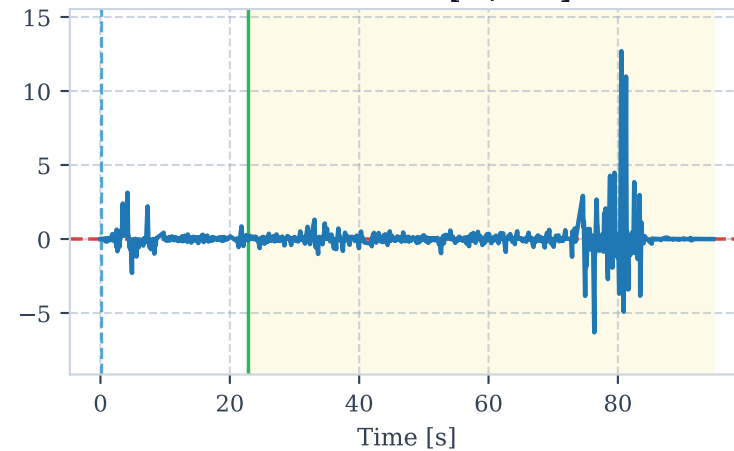
## Linear Acc X [m/s<sup>2</sup>]



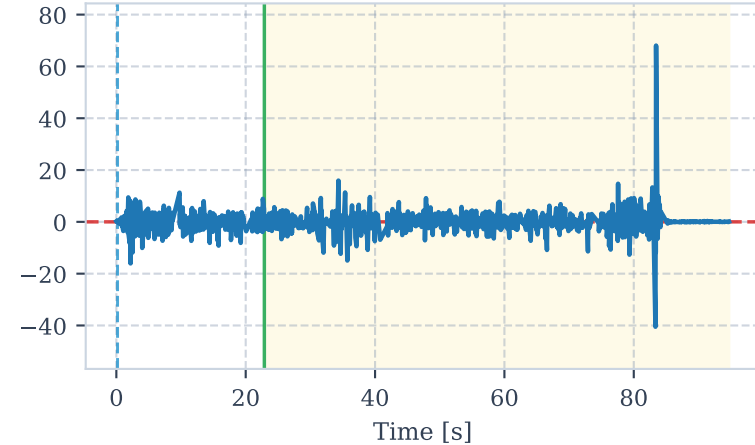
## Linear Acc Y [m/s<sup>2</sup>]



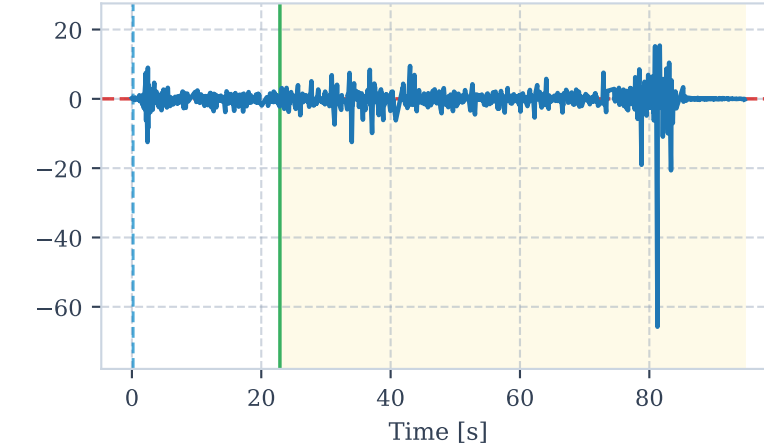
## Linear Acc Z [m/s<sup>2</sup>]



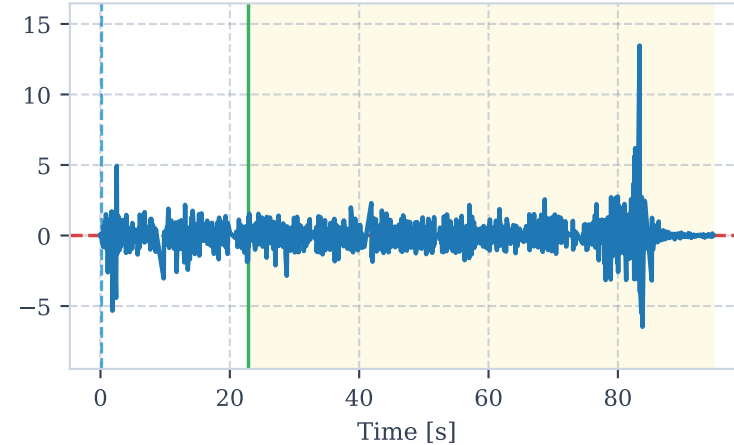
## Angular Acc X [rad/s<sup>2</sup>]



## Angular Acc Y [rad/s<sup>2</sup>]



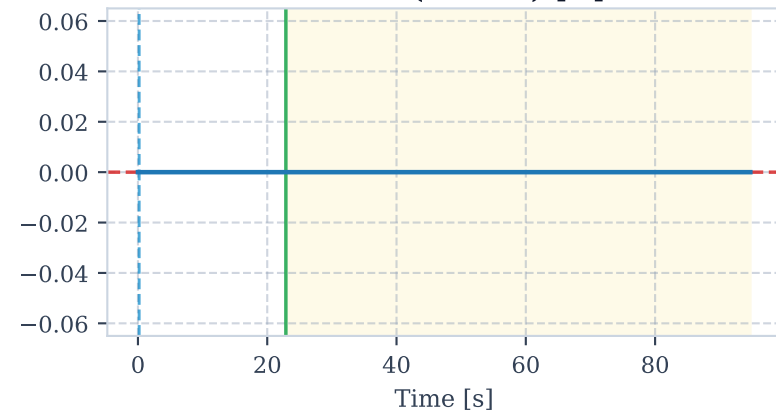
## Angular Acc Z [rad/s<sup>2</sup>]



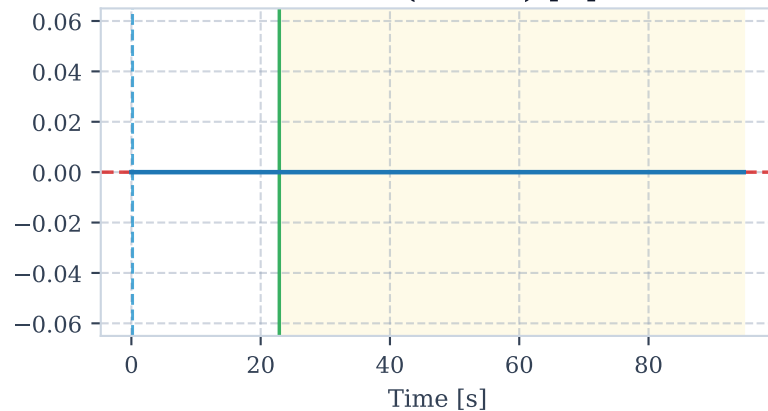
# Peg External Contact Forces (FT Sensor)



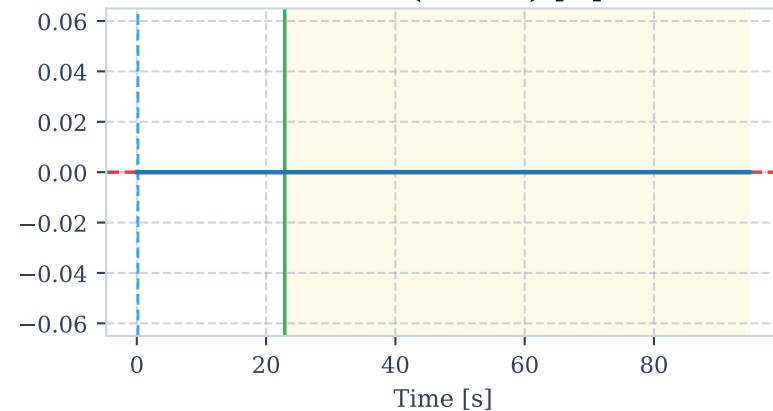
## Force X (Sensor) [N]



## Force Y (Sensor) [N]

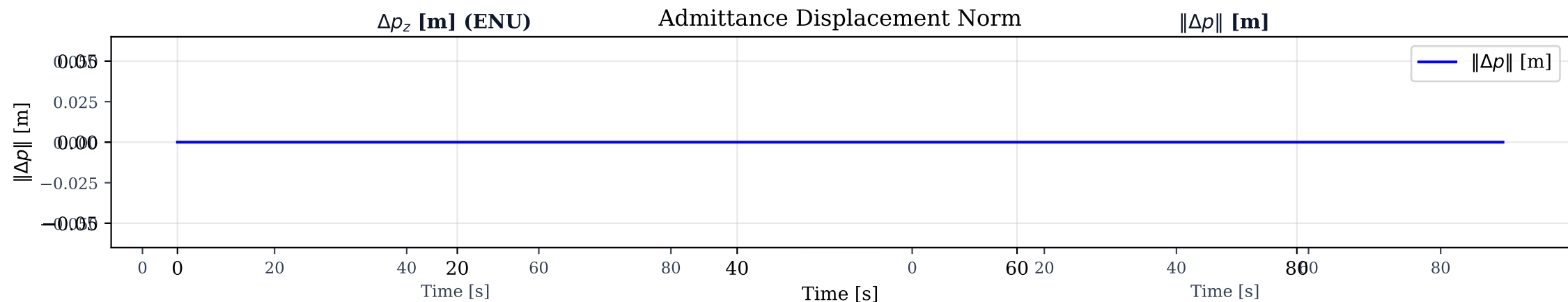


## Force Z (Sensor) [N]

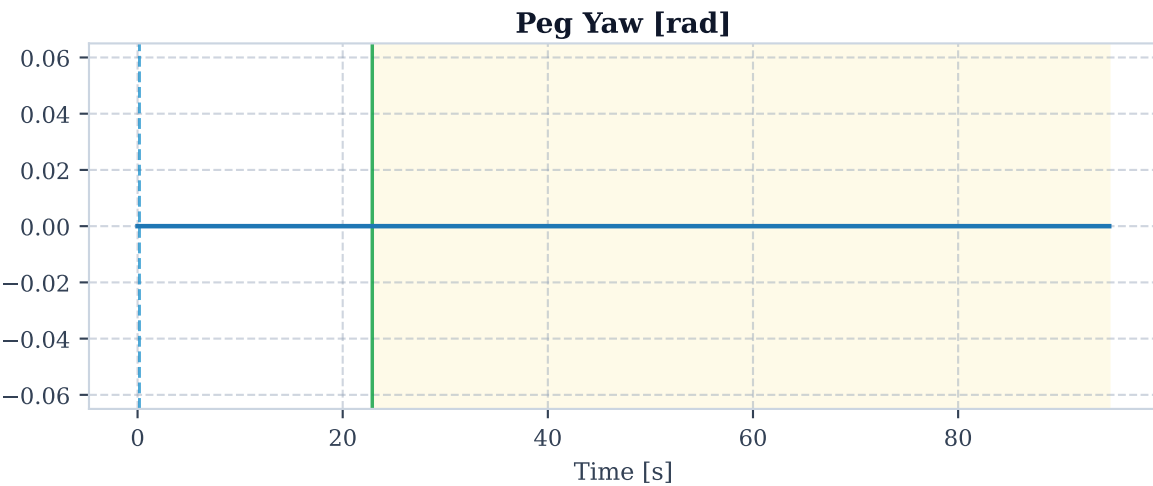
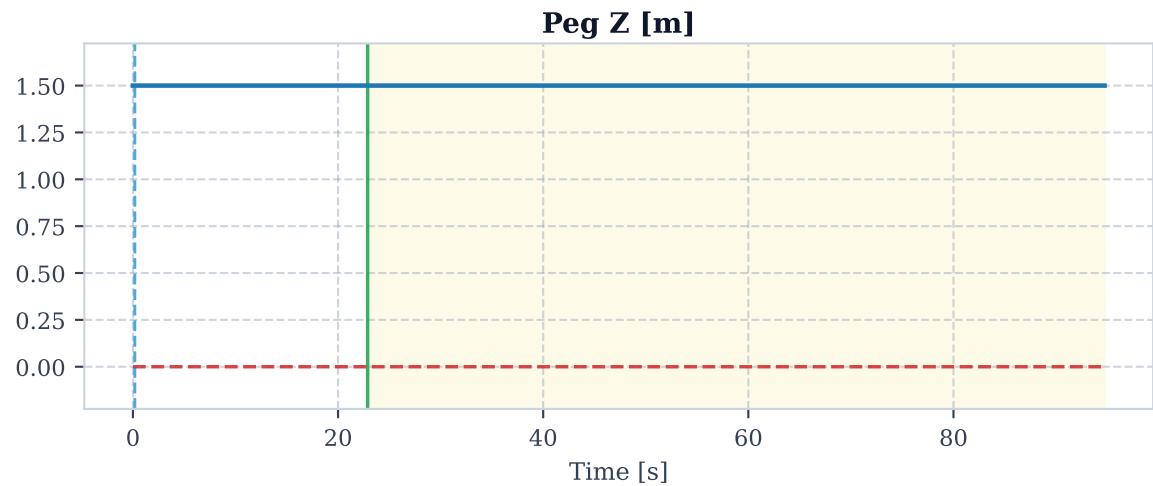
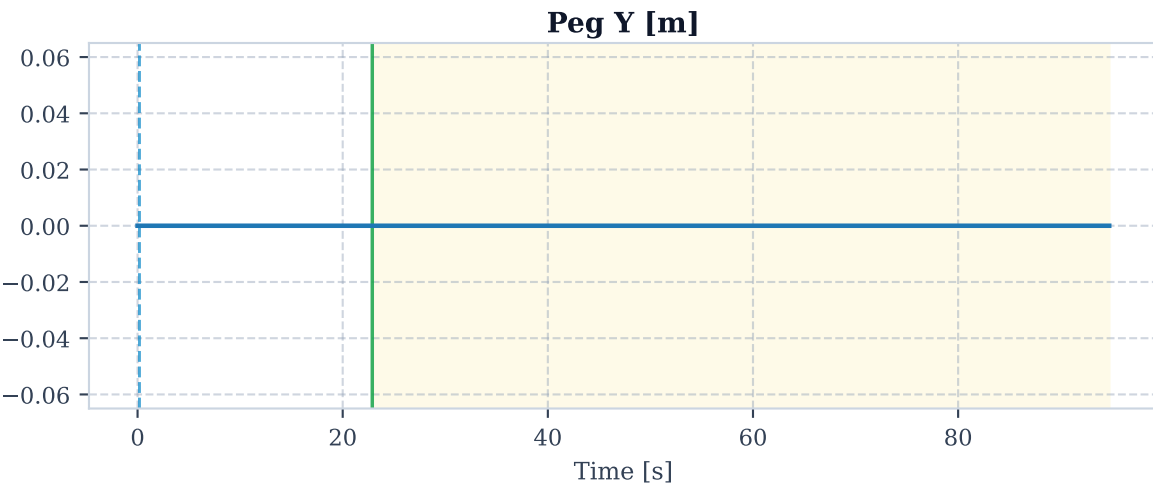
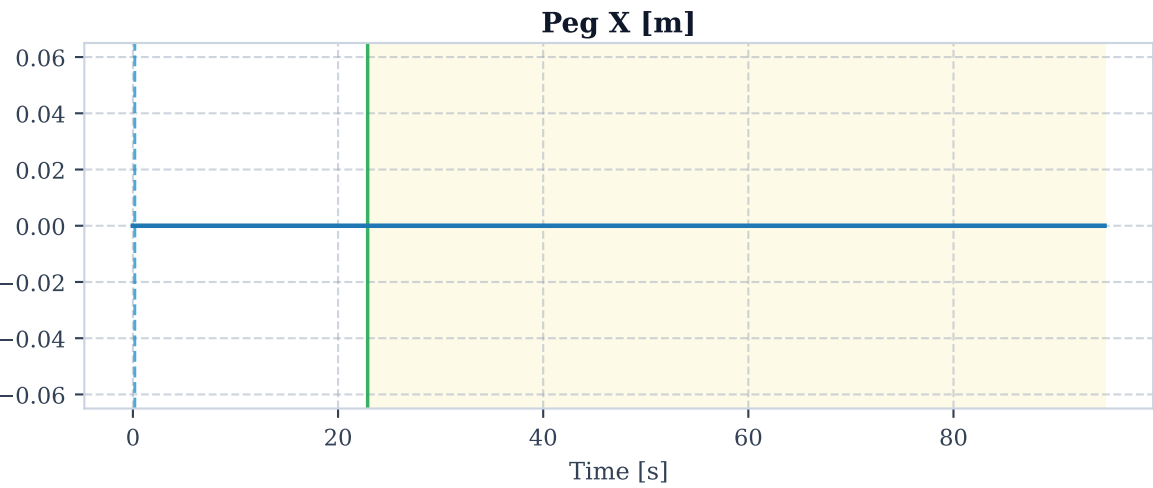


# Admittance Effect: Contact Force vs Position Deviation

MPC Active   Actual   Ref   Takeoff   Mission

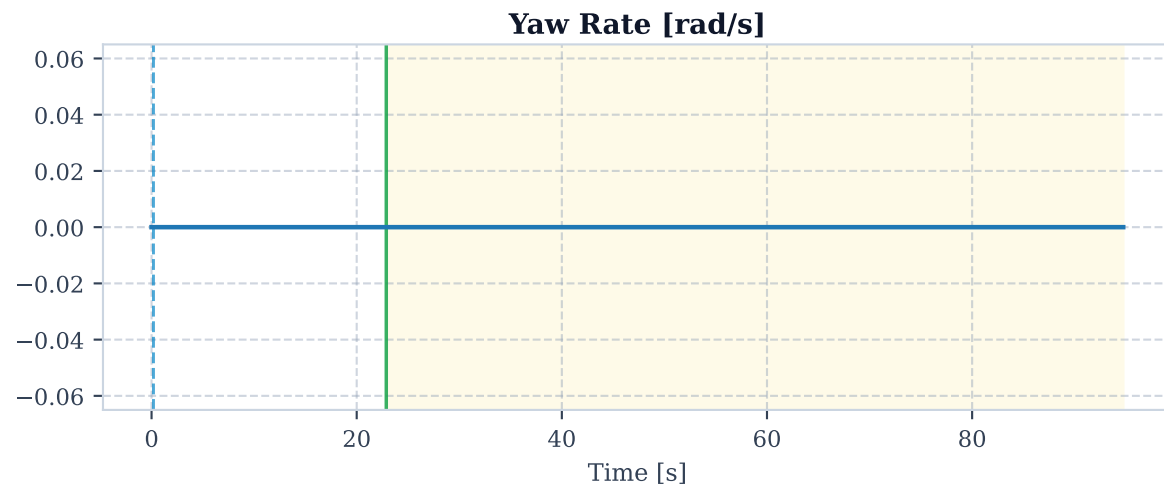
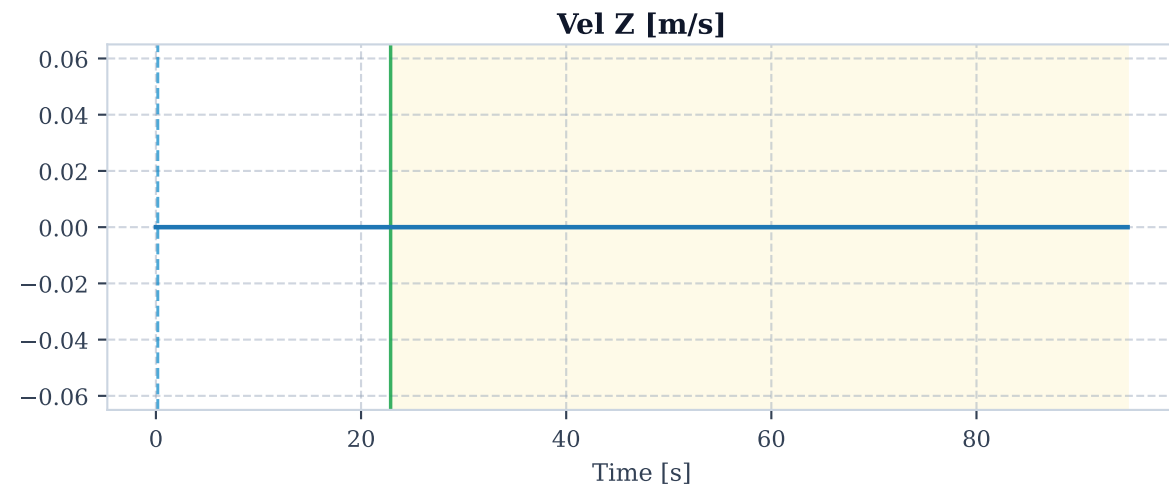
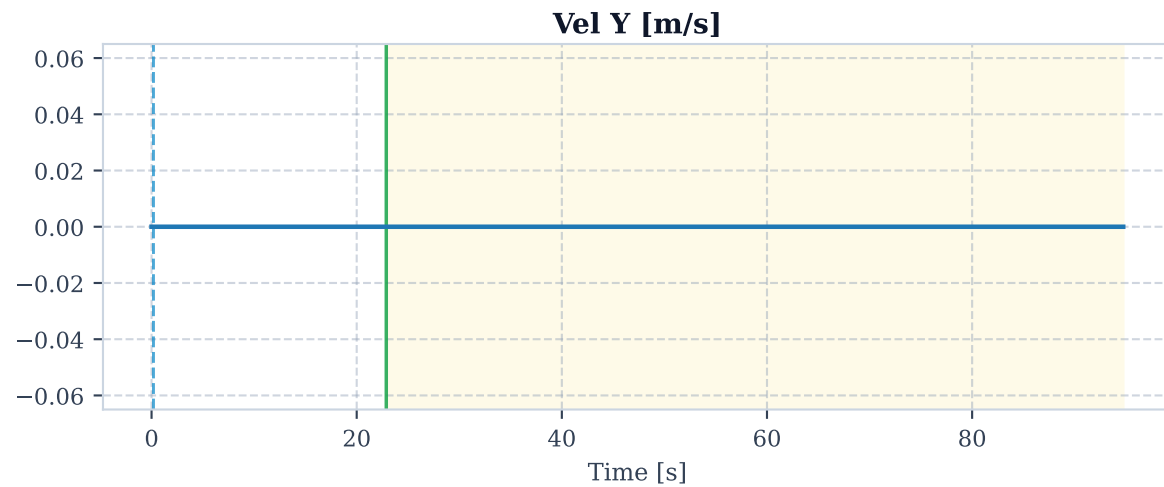
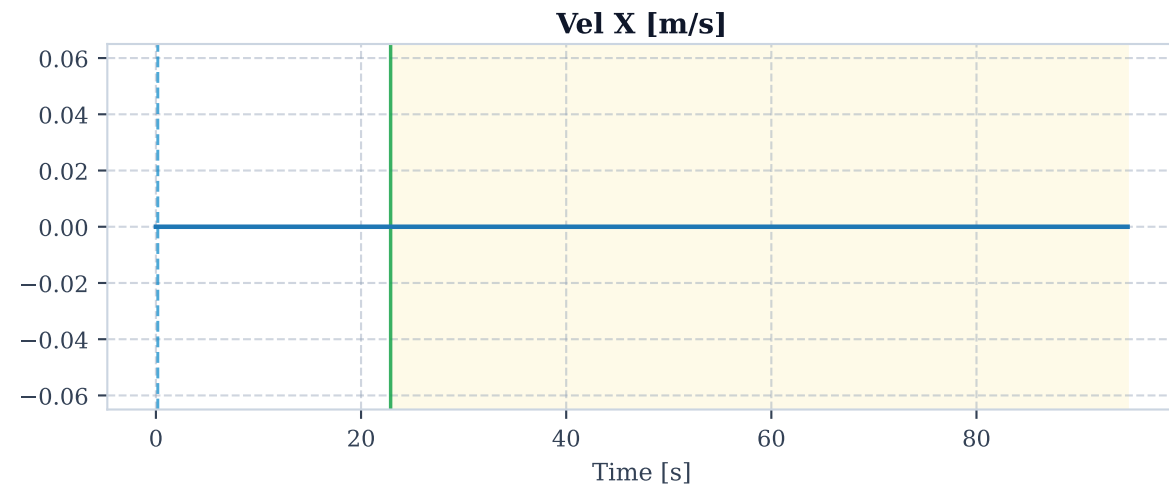


# Interaction Drone Position ENU (Actual vs Planner Reference)



# Interaction Drone Velocities (ENU) and Yaw Rate

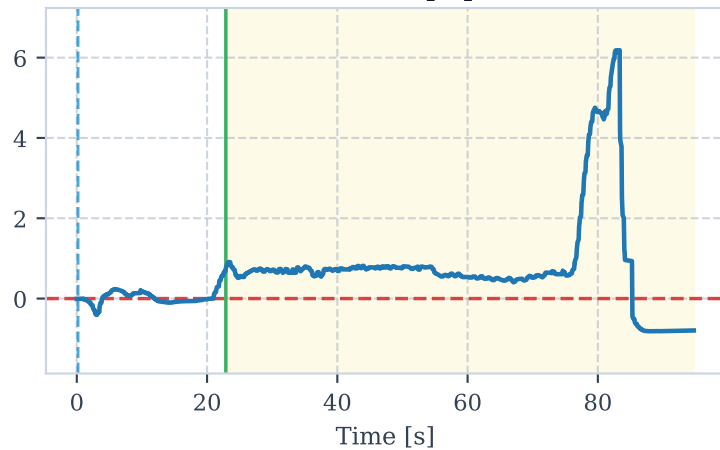
MPC Active    Actual    Reference    Takeoff    Mission



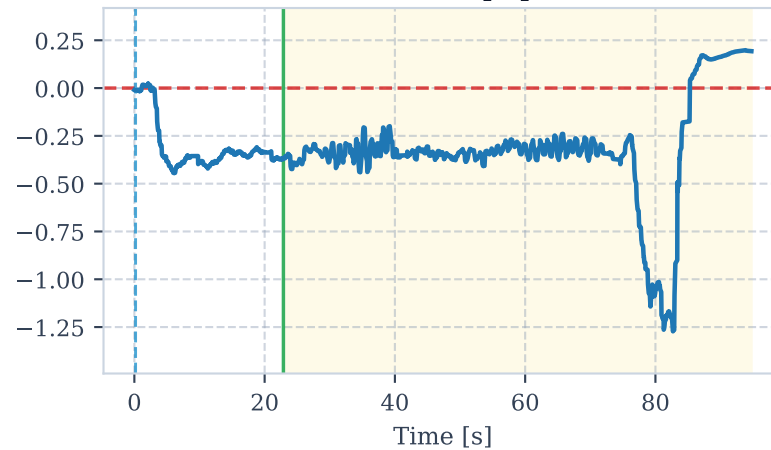
# Estimated Wrench (Momentum-Based Estimator)



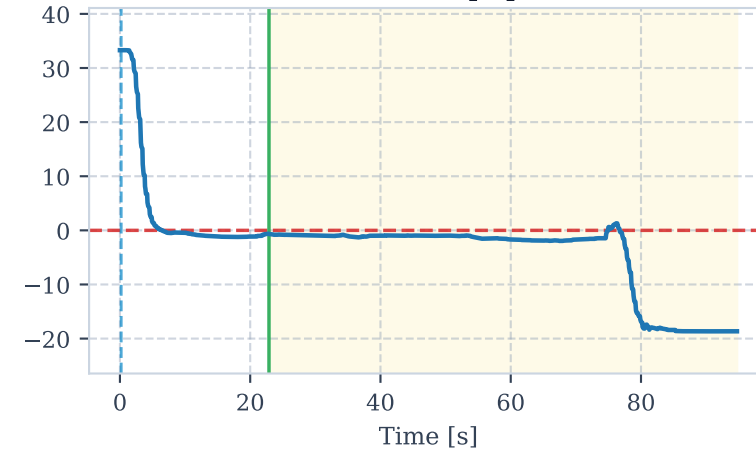
### Force X [N]



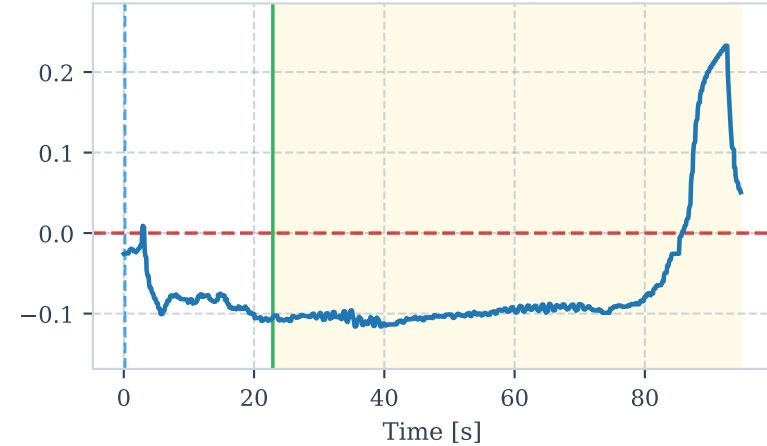
### Force Y [N]



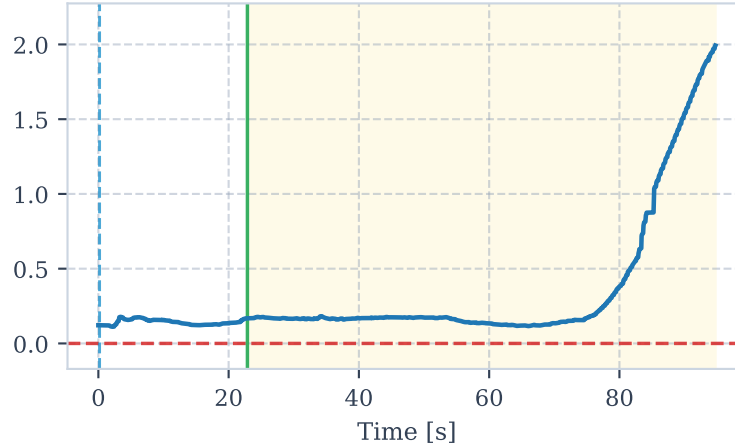
### Force Z [N]



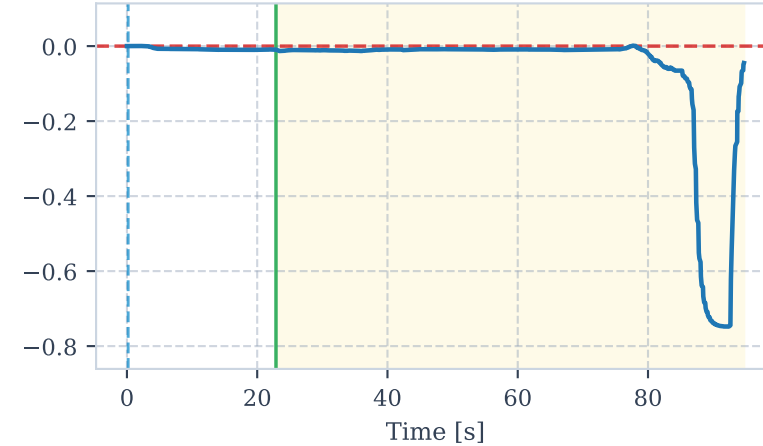
### Torque X [Nm]



### Torque Y [Nm]



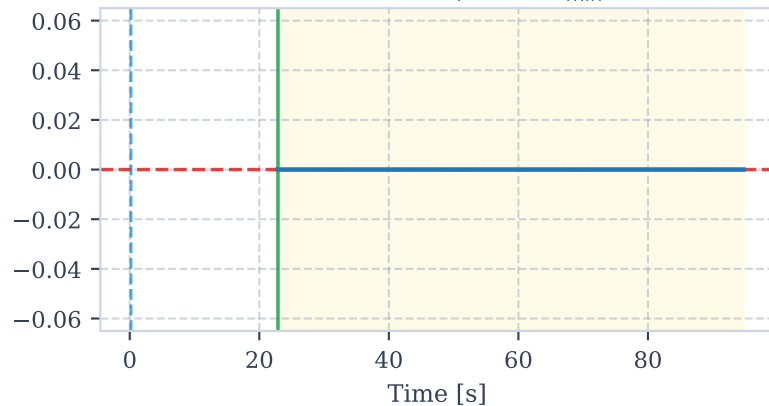
### Torque Z [Nm]



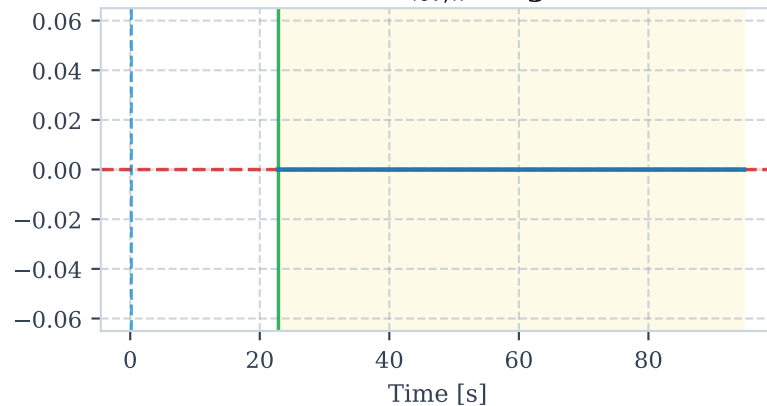
# Safety Distance and Camera FoV Constraint Violations



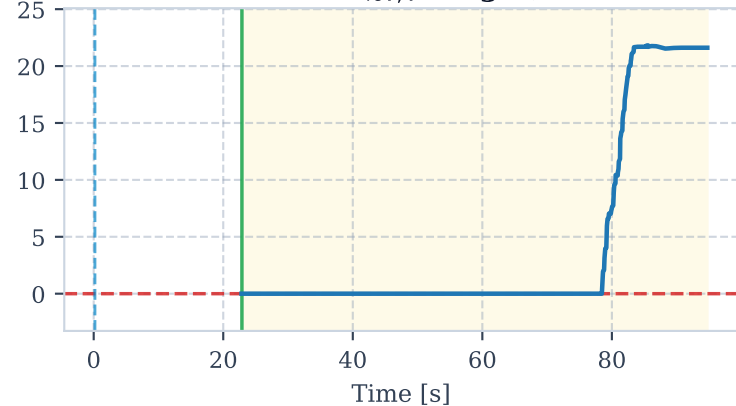
**Distance Violation  $s_r$  [m] ( $r_{min}=1.0$  m)**



**Horiz. FoV Violation  $s_{fov,h}$  [deg] (half-FoV=34°)**



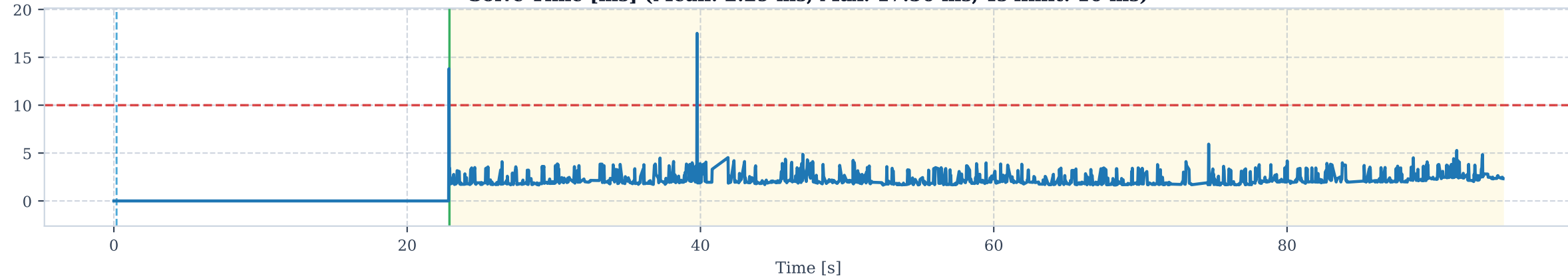
**Vert. FoV Violation  $s_{fov,v}$  [deg] (half-FoV=26°)**



# MPC Solver Execution Time & Timing Jitter



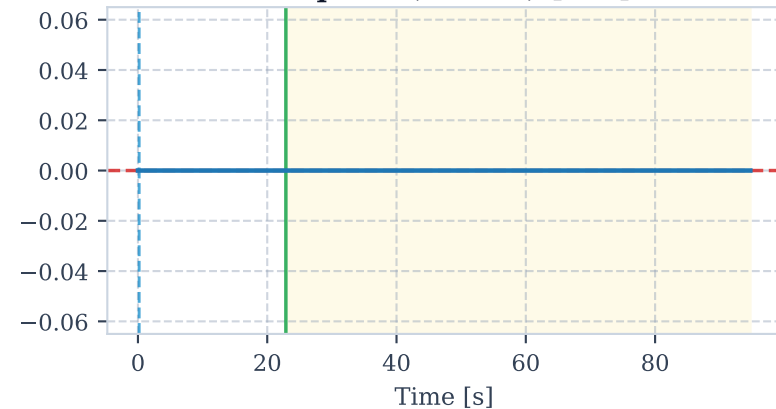
**Solve Time [ms] (Mean: 2.29 ms, Max: 17.50 ms, Ts limit: 10 ms)**



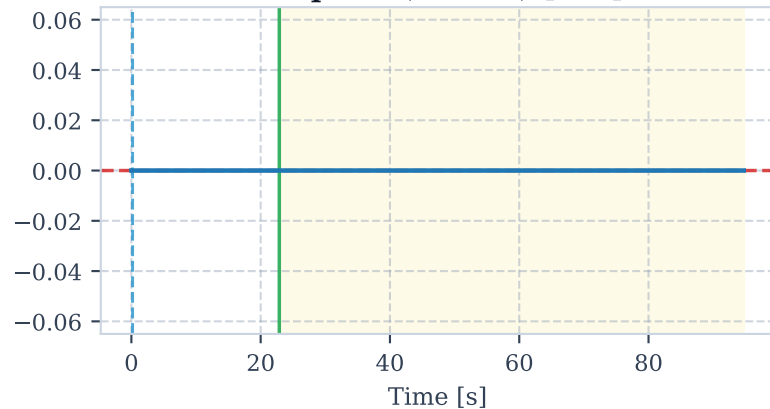
# Peg External Contact Torques (FT Sensor)



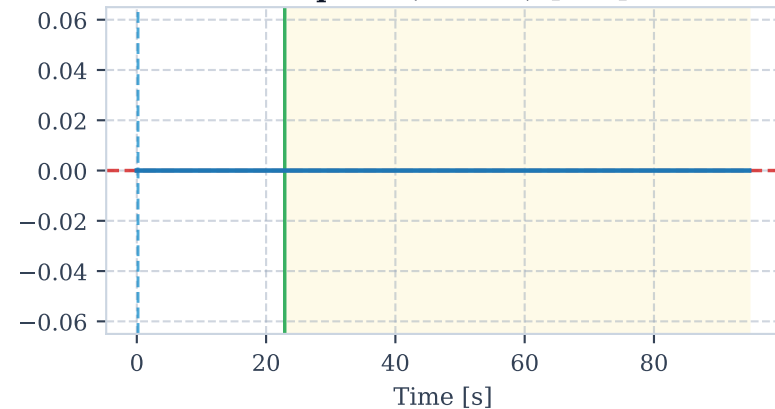
## Torque X (Sensor) [Nm]



## Torque Y (Sensor) [Nm]



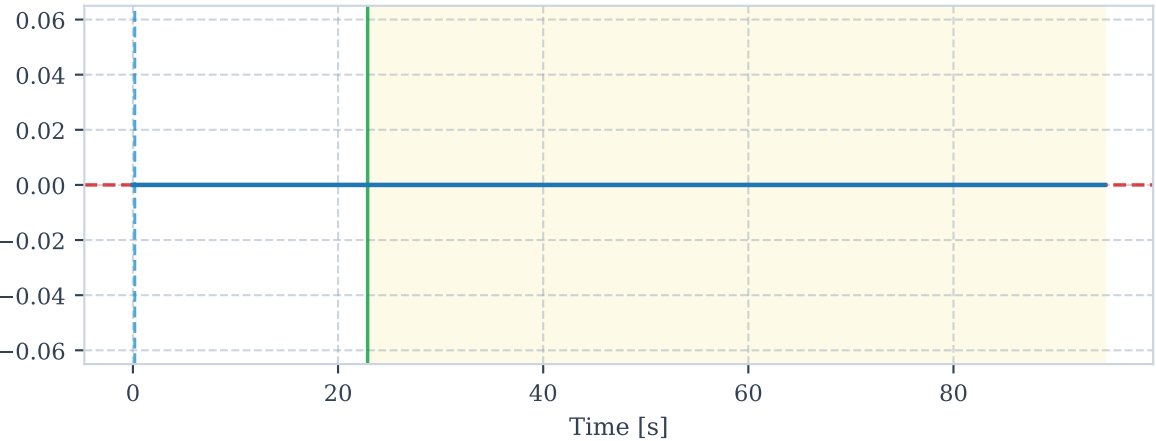
## Torque Z (Sensor) [Nm]



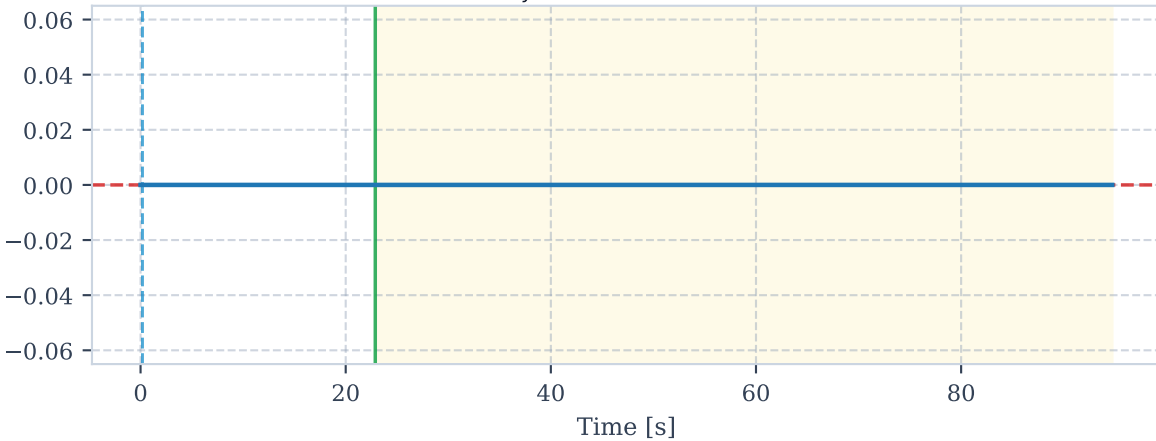
# Admittance Displacement in Sensor Frame



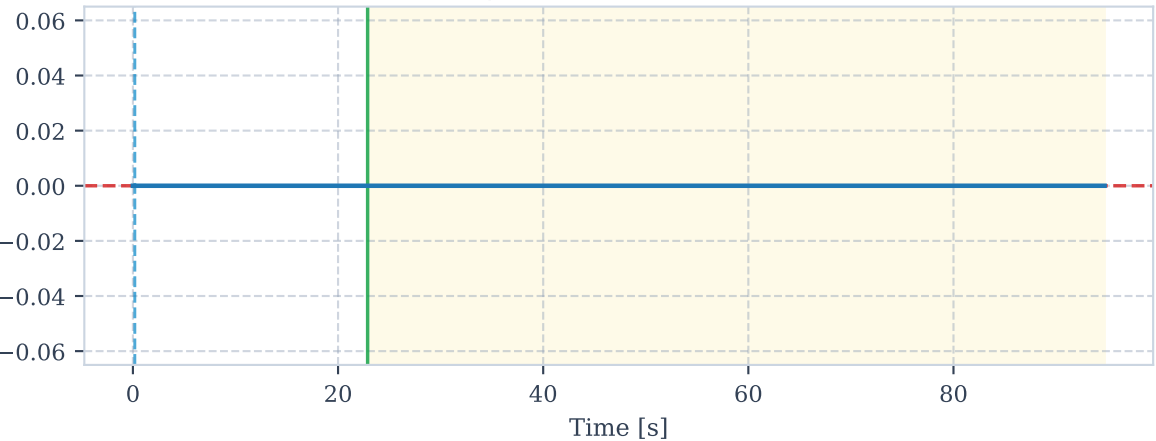
$\Delta p_{sx}$  [m] (Sensor X)



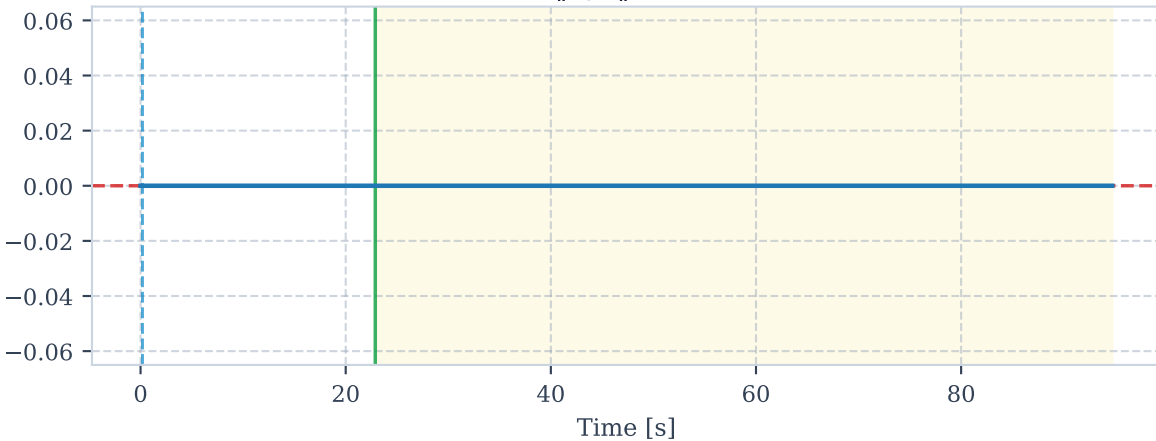
$\Delta p_{sy}$  [m] (Sensor Y)



$\Delta p_{sz}$  [m] (Sensor Z)



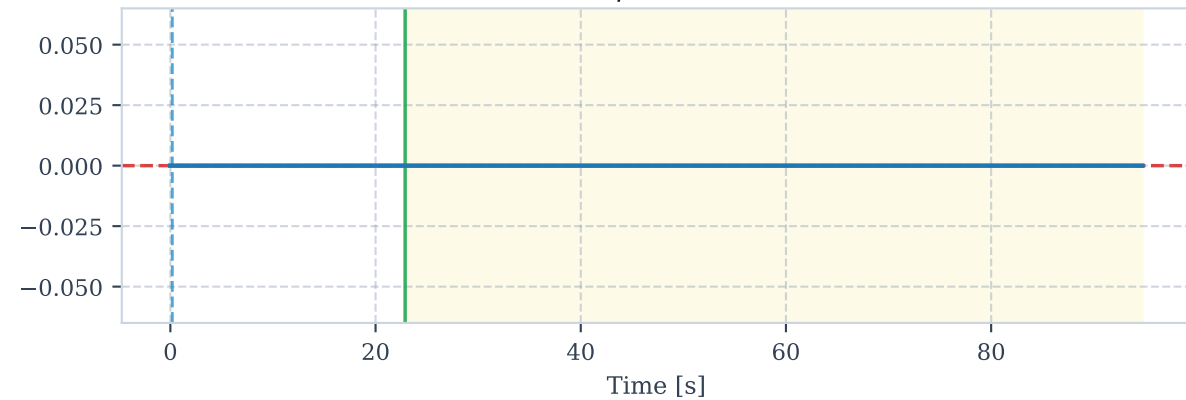
$\|\Delta p_s\|$  [m]



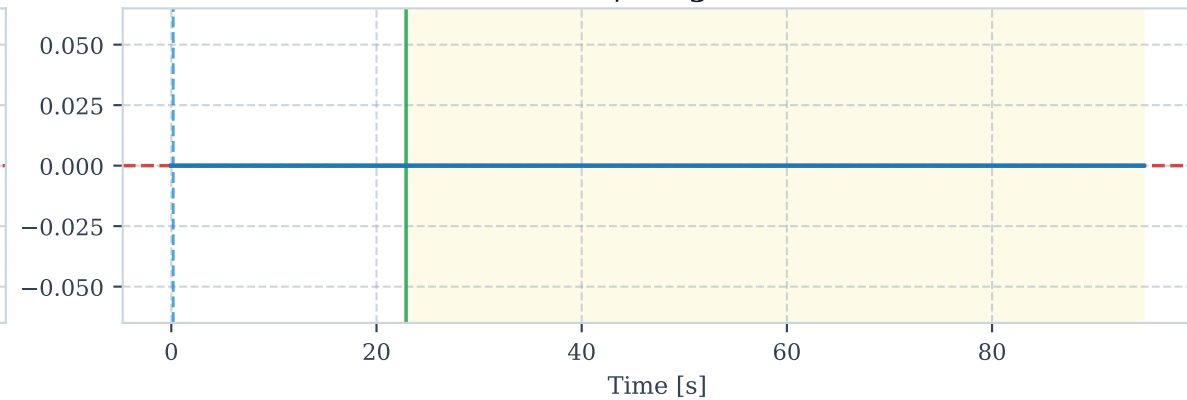
# Admittance Yaw Displacement $\Delta\psi$



$\Delta\psi$  [rad]

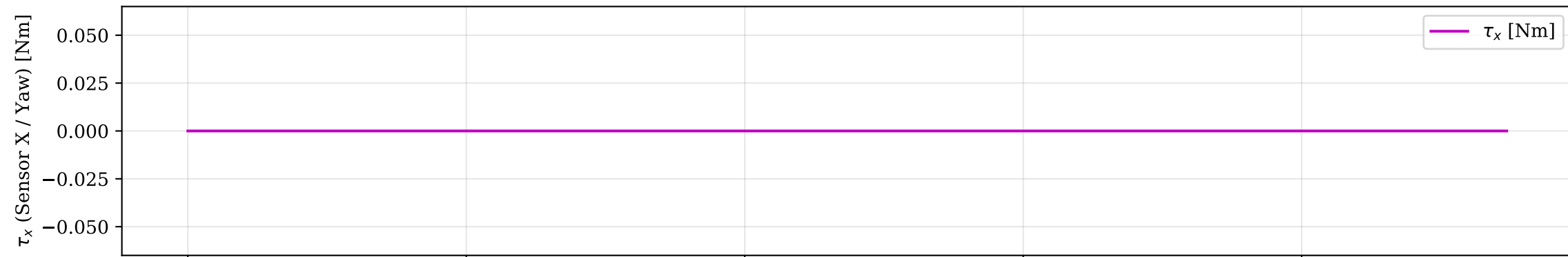


$\Delta\psi$  [deg]



# Rotational Admittance Effect: Contact Torque $\tau_x$ vs $\Delta\psi$

Contact Torque  $\tau_x$  (Sensor X / Drone Yaw axis)



Admittance Yaw Deviation  $\Delta\psi$

